

Kähler Manifolds

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Abstract

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1 Kahler Manifolds

1.1 Equivalent Definitions

Definition 1.1.1 (Kähler Forms)

Let (M, h) be a complex Hermitian manifold. Let ω be the associated form. We say that ω is a Kähler form if $d\omega = 0$. In this case, we say that M is a Kähler manifold.

Lemma 1.1.2

Let M be a Kähler manifold. Then the following are true.

- Let $g = \frac{1}{2}(h + \bar{h})$ be the associated Riemannian metric. Then (M, g) is a Riemannian manifold.
- Let $\omega = \frac{i}{2}(h - \bar{h})$ be the associated $(1, 1)$ -form. Then (M, ω) is a symplectic manifold.

Proposition 1.1.3

Let (M, ω) be a symplectic manifold. Then ω is a Kähler form if and only if M admits an integrable almost complex structure J . In this case, the following are true.

- The Riemannian metric is given by $g(u, v) = \omega(u, Jv)$.
- The Hermitian metric is given by $h = g - i\omega$.

Theorem 1.1.4 (Banyaga)

Let M be a compact complex manifold. Let ω_1 and ω_2 be Kähler forms on M . Then (M, ω_1) and (M, ω_2) are symplectomorphic.

Proposition 1.1.5

Let (M, g) be a Riemannian manifold of dimension $2n$. Then M admits a Kähler form if and only if $\text{Hol}(\nabla^g) \leq U(n)$.

Example 1.1.6

$\mathbb{CP}^n$ together with the Fubini-Study metric is a Kähler manifold.

Example 1.1.7

Every Riemann surface is a Kähler manifold.

1.2 The Hodge Decomposition and the Hodge Filtration

Theorem 1.2.1 (Hodge Decomposition Theorem)

Let M be a compact Kähler manifold. Then the following are true.

- There is an isomorphism

$$H_{\text{dR}}^k(M; \mathbb{C}) \cong \bigoplus_{p+q=k} H^{p,q}(M)$$

for all $k \in \mathbb{N}$.

- There is an isomorphism

$$H^{p,q}(M) \cong \overline{H^{q,p}(M)}$$

for all $p, q \in \mathbb{N}$.

- $\dim_{\mathbb{R}}(H_{\text{dR}}^k(M))$ is even for all $k \in \mathbb{N}$.

Example 1.2.2

Every compact Riemann surface is of the form Σ_g for some $g \geq 0$. We have seen above that Σ_g is

Kähler. We know that

$$H_{\text{dR}}^k(\Sigma_g; \mathbb{C}) = \begin{cases} \mathbb{C} & \text{if } k = 0, 2 \\ \mathbb{C}^{2g} & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

From this we conclude that $H^{0,0}(\Sigma_g) = \mathbb{C}$. Now since $\dim_{\mathbb{C}}(H^{p,q}(M)) = \dim_{\mathbb{C}}(\overline{H^{q,p}(\Sigma_g)}) = \dim_{\mathbb{C}}(H^{q,p}(\Sigma_g))$, we conclude that $H^{1,0}(\Sigma_g) \cong H^{0,1}(\Sigma_g) \cong \mathbb{C}^g$. Also, since Σ_g has complex dimension 1, it has no holomorphic 2-forms. Hence $H^{2,0}(\Sigma_g) = H^{0,2}(\Sigma_g) = 0$. This leaves us with $H^{1,1}(\Sigma_g) = H^2(\Sigma_g) = \mathbb{C}$. The hodge diamond of Σ_g is given by

$$\begin{array}{ccc} & 1 & \\ g & & g \\ & 1 & \end{array}$$

Definition 1.2.3 (Hodge Filtration)

Let M be a complex manifold. Define the Hodge filtration by

$$(F^i \Omega_M^\bullet)^j = \begin{cases} \Omega_M^j & \text{if } j \geq i \\ 0 & \text{otherwise} \end{cases}$$

Define the Hodge filtration on cohomology by

$$F^i H^k(M, \mathbb{C}) = \text{im} (\mathbb{H}^k(M, F^p \Omega_M^\bullet) \rightarrow \mathbb{H}^k(M, \Omega_M^\bullet) \cong H^k(M, \mathbb{C}))$$

The hypercohomology of a complex of sheaves always admits a double complex, which then gives us a spectral sequence.

Definition 1.2.4 (Analytic Hodge de Rham Spectral Sequence)

Let M be a complex manifold. Define the Hodge de Rham spectral sequence to be the spectral sequence associated to the Hodge filtration $F^\bullet \Omega_M^\bullet$.

Proposition 1.2.5

Let M be a compact Kähler manifold. Then we have $E_1^{p,q} = H^{p,q}(M) \Rightarrow H^{p+q}(M)$.

Corollary 1.2.6

Let M be a compact Kähler manifold. Then we have

$$F^i H^k(M; \mathbb{C}) = \bigoplus_{p \geq i} H^{p,k-p}(M)$$

1.3 The Lefschetz Operator

Definition 1.3.1 (The Lefschetz Operator)

Let M be a Kähler manifold. Let ω be its associated form. Define the Lefschetz operator to be the map $L : \Omega^k(M) \rightarrow \Omega^{k+2}(M)$ given by

$$\alpha \mapsto \alpha \wedge \omega$$

Definition 1.3.2 (Adjoint of the Lefschetz Operator)

Let M be a Kähler manifold. Define the adjoint of L to be the map $\Lambda : \Omega^k(M) \rightarrow \Omega^{k-2}(M)$ such that

$$\langle \Lambda \omega, \eta \rangle_{L^2} = \langle \omega, L \eta \rangle_{L^2}$$

for all $\omega \in \Omega^k(M)$ and $\eta \in \Omega^{k-2}(M)$.

Lemma 1.3.3

Let M be a Kähler manifold. Then we have

$$\Lambda = *^{-1} L *$$

Proposition 1.3.4 (The Kähler Identities)

Let M be a Kähler manifold. Then the following are true.

- $[\Lambda, \bar{\partial}] = -i\partial^*$.
- $[\Lambda, \partial] = -i\bar{\partial}^*$.

Theorem 1.3.5 (Hard Lefschetz Theorem)

Let M be a compact Kähler manifold. Then the following are true.

- The Lefschetz operator induces an isomorphism

$$L^k : H_{\text{dR}}^{n-k}(M) \xrightarrow{\cong} H_{\text{dR}}^{n+k}(M)$$

Definition 1.3.6 (Primitive Cohomology)

Let M be a compact Kähler manifold. Let $n = \dim(M)$. Define the primitive cohomology of M to be

$$H_{\text{prim}}^i(M) = \ker(L^{n-i+1} : H_{\text{dR}}^i(M) \rightarrow H_{\text{dR}}^{2n-i+2}(M))$$

Proposition 1.3.7 (Lefschetz Decomposition)

Let M be a compact Kähler manifold. Then there is an isomorphism

$$H_{\text{dR}}^k(M) = \bigoplus_{2r \leq k} L^r (H_{\text{prim}}^{k-2r}(M))$$

Theorem 1.3.8 (The Hodge Index Theorem)

Let M be a compact Kähler manifold of complex dimension $2n$. Suppose that

$$h = \sum_{p=0}^{2m} \sum_{q=0}^{2m} (-1)^p \dim_{\mathbb{C}}(H^{p,q}(M))$$

Then the signature of the non-degenerate symmetric bilinear form $H^n(X; \mathbb{R}) \times H^n(X; \mathbb{R}) \rightarrow \mathbb{R}$ from Poincaré duality is given by $(h, 2n - h)$.